Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/13/2007
PI Name:	Pollack, Lois Ph.D.		
Project Title:	Microscale Mixer for Protein Folding		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Batt, Carl (Cornell University) Batt, Carl (Cornell University)		
Grant/Contract No.:	NAG3-2942		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the un iverse obscured by Earth's gravity		
Task Description:	Biological macromolecules, like proteins, DNA's and RNA's, self-assemble or fold into specific nano-meter sized structures that enable life. The broad, long-term objective of this work is to understand the interactions involved in macromolecular folding and function. To observe these interesting dynamics, it is necessary to trigger and monitor structural changes on time scales as short as microseconds. Our present understanding is limited by the lack of suitable experimental tools capable of reporting dynamics on time scales that range from microseconds to seconds. For example, few instruments can follow the rapid changes in the most basic of molecular parameters: size and shape. We have applied synchrotron small angle x-ray scattering (SAXS) techniques to probe the folding and function of both DNA and RNA. This funding facilitated the development of two enabling technologies for x-ray scattering studies of macromolecular systems: a) the use of ultra-thin silicon nitride films as windows for microfabricated cells and b) the		

use of glass capillaries to focus x-rays in SAXS experiments. These new techniques enabled studies of RNA folding, in addition to measurements quantifying the distribution of positive ions around (negatively-charged) DNA and interactions between DNA's.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This research focused on developing new technologies that enabled measurements in hard-to-access regimes. In addition to making these technologies available (through information provided in publications), the funding also partially supported research into biological systems. Our focus is on describing the fundamental physics of DNA and RNA. These molecules are critical for life on earth.

Task Progress:

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1. Development of silicon nitride windows for use with small volume, microfabricated cells for x-ray scattering experiments. These windows have enabled measurements of RNA folding on time scales ranging from 1 ms through 10 minutes, as well as measurements of the distribution of counterions around DNA and interactions between DNA's.

To carry out SAXS experiments on small samples, a great deal of effort must be expended in decreasing the noise due to background scattering. Most of the biological samples of interest to this project consist of macromolecules in solution, thus, they must be contained. Sample cells must be sealed with windows that pass x-rays with minimal attenuation. Furthermore, it is difficult to construct refillable and vacuum compatible sample, so it is often necessary to break the vacuum through which the x-rays pass. In this case, windows must be inserted along the path of the x-ray beam. Overall, x-rays in a typical scattering experiment, must pass through four or five windows; each adds to the background scatter. This high background is especially problematic for samples that scatter weakly, such as our molecules within microfabricated flow cells. Our signal is easily dominated by background scattering from the windows. This program enabled the fabrication and use of freestanding silicon nitride membranes as x-ray windows for both our sample cells and vacuum windows. These membranes range from 200nm to 1μm in thickness. They present low and reproducible scattering backgrounds for our experiments. These windows were fabricated at the Cornell Nanoscale Facility (CNF).

These windows have been employed to seal microfabricated rapid fluid mixers, as well as specially designed cuvettes for stopped flow mixers. Taken together, these technologies enable measurement times spanning form 1 ms through 10 minutes. Our measurements on RNA folding indicate that the first step of in vitro experiments corresponds to a rapid compaction or collapse of the molecules, following the addition of the small, positively charged ions that trigger the folding [1, 2]. This observation highlights the importance of the diffuse cloud of small ions that surrounds nucleic acids and has led to a second series of experiments probing the equilibrium distribution of cations around negatively charged DNA [3, 4]. The goal of these experiments is to determine the spatial distribution of ions as a function of increasing ion charge.

Only by gaining a detailed understanding macromolecular folding and function on earth, can we begin to understand the molecular mechanisms for changes resulting from space travel. An unfortunate by-product of space travel is the ubiquitous, high level of background radiation. It is well known that DNA is susceptible to radiation damage. At least one mechanism for damage involves interactions with the solution that immediately surrounds the DNA, once the entire system has been exposed. It is therefore important to have an accurate representation of DNA's solution environment. The use of silicon nitride windows facilitated measurements of the competition of monovalent and di valent ions for screening of DNA's negative charge and enabled measurement of the ion concentration at the surface of the DNA [3].

Finally, the use of these windows enabled studies of interactions between short DNA strands [5, 6]. Small changes in the SAXS profiles measured at the lowest scattering angles report on interactions between neighboring DNA strands. These interactions can be tuned from repulsive through attractive, by varying the valence and concentration of counterions in solution. Because the changes can be small, it is essential to reduce all extraneous sources of noise, silicon nitride windows used along the x-ray beamline and for the sample cells provide the requisite low noise background.

2. The use of focusing capillary optics for solution small angle x-ray scattering experiments.

A second technological advancement made possible by this funding is the use of borosilicate glass focusing capillaries for SAXS experiments [7]. Single-bounce focusing capillaries can be used in conjunction with synchrotron x-rays to create high flux microbeams. With these capillaries, we achieve high x-ray flux in a spot size of tens of microns. This capability is ideal for use in conjunction with microfluidic mixers where sample volumes are small and time resolution is determined by beam size. SAXS experiments carried out on a model protein system show no deleterious effects result from the use of these focused beams [7].

References—all publications contain an acknowledgement to NASA for funding--listed below in order referenced in the above document.

1. Das, R., L.W. Kwok, I.S. Millett, Y. Bai, T.T. Mills, J. Jacob, G.S. Maskel, S. Seifert, S.G.J. Mochrie, P. Thiyagarajan, S. Doniach, L. Pollack, and D. Herschlag, The fastest global events in RNA folding: Electrostatic relaxation and tertiary collapse of the tetrahymena ribozyme. Journal of Molecular Biology, 2003. 332(2): p. 311-319.

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3. Andresen, K., R. Das, H.Y. Park, H. Smith, L.W. Kwok, J.S. Lamb, E.J. Kirkland, D. Herschlag, K.D. Finkelstein, and L. Pollack, Spatial distribution of competing ions around DNA in solution. Physical Review Letters, 2004. 93(24).

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Bibliography Type:	Description: (Last Updated: 07/19/2007)
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Articles in Peer-reviewed Journals	Kwok LW, Shcherbakova I, Lamb JS, Park HY, Andresen K, Smith H, Brenowitz M, Pollack L. "Concordant exploration of the kinetics of RNA folding from global and local perspectives. " J Mol Biol. 2006 Jan 13;355(2):282-93. Epub 2005 Nov 10. PMID: 16303138 , Jan-2006
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